USER MANUAL

Six-Layer Model Geometry Visualizer

Two interactive web panels that render a six-layer state-space model as three-dimensional geometry. You adjust the parameters and observe, in real time, how they affect the model's state and its diagnostic quantities.

Technical documentation · for academic use · not investment advice and not a market forecast.

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1. Overview

The application consists of **two independent HTML pages**, each a self-contained control panel (no server, no installation required). Both render with WebGL and run all computation directly in the browser.

- **Panel A** focuses on a single mechanism: the influence of one layer on another is split into two components — a *tangential* one (which displaces a point on a layer surface) and a *normal* one (which displaces the layer surface itself). A single parameter, τ, controls the ratio between them.
- **Panel B** shows the full manifold: all six layer surfaces $\Sigma_0 \ldots \Sigma_5$ together with the normal axis $\perp$, the state of each layer, the seven inter-layer coupling edges, and the seventeen channels that contribute to the output.

The model keeps all **78 variables** intact, organized by the *geometric role* of each weight. There are four roles:

Role	Geometric meaning
Position	Displaces the layer surface itself (the normal component acts on this role).
Tangential	Displaces a point on the layer surface without moving the surface.
Curvature	How much the layer surface bends.
Amplitude	The maximum offset the layer surface can reach.

At `τ = 0` the influence lies entirely in the tangential component; the model coincides with its baseline form and every diagnostic quantity is zero to machine precision. When `τ > 0`, part of the influence moves into the normal component and produces a *persistent displacement* (denoted `ξ`) — the part of the motion that does not reverse when the influence is removed.

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2. System requirements and launching

- A modern browser with **WebGL** support: a recent Chrome, Edge, Firefox or Safari.
- No network connection is needed for computation; a connection is used only to load the fonts and the rendering library (if offline, the page still runs with fallback fonts).
- **To launch:** double-click the HTML file to open it in a browser, or place the file on a web server and visit the corresponding URL.

The interface switches automatically between light and dark appearance according to the operating-system setting.

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3. Common 3-D viewport controls

Both panels contain an interactive three-dimensional viewport:

- **Drag** (mouse or finger): rotate the scene around its center.
- **Scroll** (or pinch): zoom in and out.

Every change in the control block is applied to the viewport and the numeric readouts immediately; there is no confirm button.

4. Panel A — Tangential and normal decomposition

This panel illustrates one mechanism on a pair of layers: it holds the two events and the final state fixed, varies only the normal fraction and the order, and reads back three diagnostic quantities.

4.1. Control block

Control	Range	Function
τ_{45} slider normal fraction	`0 → ~0.262 (15 steps)`	`τ = 0`: fully tangential influence (changes coefficients on the surface only). Increasing `τ`: moves part of it into the normal component (displaces the surface itself). The current value is shown next to the label. **The usable region ends at ≈ 0.262**; steps beyond the cap can still be dragged but raise an *"out of range"* warning to show why one should not exceed it.
Layer-4 shock amplitude	`0.60 → 1.00 step 0.02`	Sets a common intensity for the Layer-4 input group. The threshold that activates the normal component is `H* = 0.45`, evaluated on the layer's aggregate quantity rather than directly on this amplitude.
Order of the two events	2 toggle buttons	Selects the sequence of the two events (button `meme → cứng` or `cứng → meme`). The two events and the final state are identical in both choices; any remaining difference can only come from the order.
Restore button	Toggle	At the bottom of the control block. Press it to remove the influence and return the observed state to its pre-shock value; any remaining *persistent displacement* `ξ` is kept visible.

4.2. Quick labels on the viewport

The corner of the viewport shows four instantaneous status labels:

- A **tangential-component indicator** and a **normal-component indicator** — lit according to the current `τ`.
- `Λ` — the logistic value at the current operating point.
- `ξ₅` — the persistent displacement accumulated on layer `Σ₅`.

4.3. The three diagnostic quantities

At `τ = 0` all three are zero — not because they were calibrated to zero, but because the output equation is a function of state and carries no history: two paths to the same state necessarily give the same result.

Quantity	Meaning
$\|[L_4, L_5]\|$ commutator	Measures the difference when the order of the two events is reversed. Zero means the order does not affect the result.
$H(\gamma)$ holonomy	Traverses a closed loop in parameter space and returns to the starting point. Non-zero means the loop leaves behind a persistent displacement.
residual after unwinding	What remains after the shock is removed. Zero means the influence is fully reversible.
θ_0 final threshold	The final position of layer Σ_5. Fixed at $\tau = 0$; it may shift once $\tau > 0$.

4.4. Fragmentation-input section

The lower part of the panel illustrates a Layer-2 input variable under the rule $T_{frag}(t) = T_{frag}^0 + \rho \cdot \xi_2(t)$. Two modes are compared:

- **No feedback** — the variable keeps the value entered; the model cannot generate it on its own.
- **With feedback** — the variable gains an additional contribution from the persistent displacement ξ_2 at the current τ.

The remaining sections list the five tuning constants (baseline, usable region, provenance, evidence level) and a note on the cells the user must fill in manually. These are reference-only, read-only sections.

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5. Panel B — Six-sheet manifold

This panel gives the full picture: all six layer surfaces, the state of each layer, the coupling edges, and the channels that contribute to the output. The layout has a 3-D viewport column on the left and an input column on the right.

5.1. Readouts at the top

Symbol	Meaning
`ln DXY`	The model's scalar output quantity (a log of the index).
`Ψ_str, κ_sys, Λ(σ)`	Intermediate model quantities at the current state.
`Σ a∥ · Σ a⊥ · Σ ξ`	Total tangential component, total normal component, and total persistent displacement across the manifold (also shown as labels on the viewport).
`‖[L_a, L_b]‖ · H(γ) · residual`	The three diagnostic quantities, defined as in Panel A but computed on the layer pair under consideration.
`‖C‖∞ < 4 · η·κ_A·g₁ < 4`	Two bounds that check the model's contraction property; the values must stay below the threshold of 4.

5.2. Input column (right side)

Each variable has a numeric field and an accompanying slider, bounded to that variable's `[lo, hi]` region. The buttons at the top of the column set the entire number set at once:

Button	Effect
`Table 10.3`	Loads the baseline value set (default).
`ngẫu nhiên`	Generates a random value set within the usable region.
`mở`	Sets the variables to the “open” configuration.
`đóng`	Sets the variables to the “closed” configuration.

A pair of toggle buttons for the **memory mode** determines how persistent displacement accumulates:

- `trễ · cực đại` (hysteresis · maximum) — the displacement equals the **historical maximum**; only the peak value of the path counts, duration does not enter the formula (default).
- `mỏi · tích phân` (fatigue · integral) — the displacement is the **time integral**; duration can substitute for amplitude.

A cell marked with a distinct color denotes a variable that has *no observed series* (filled in by the user).

5.3. The three reference tables

“Six sheets” table ($\Sigma_0 \ldots \Sigma_5 \cdot \perp$)

Each row is a layer, with columns for: the position-role weight, the `[lo, hi]` region, the current value, the raw displacement, the displacement after clamping to the valid region, the intensity, and the curvature `II = 4Λ'`.

“Seven inter-layer edges” table

Each row is a coupling edge between two layers, with: source layer, destination layer and its weight, the role of the destination weight, the load, `τ`, the tangential component `a∥ = (1−τ)·load`, the normal component `a⊥ = τ·load`, the threshold excess `(src − H*)₊`, and the displacement depth `τ·p·excess·(1+q·II)`.

“Seventeen channels” table

Lists the channels contributing to the output under the rule `α(σ) = α_static + (α_taut − α_static)·Λ(σ)`, with each channel's side, level, exponent, log term and evidence level.

6. Glossary of quantities

Symbol	Name	Interpretation
τ	Normal fraction	The fraction of the inter-layer influence that lies along the normal direction. `τ = 0`: fully tangential.
a∥	Tangential component	`(1−τ)·load` — displaces a point on the surface; reversible.
a⊥	Normal component	`τ·load` — displaces the surface itself.
ξ	Persistent displacement	The non-reversing part of the motion; monotonically non-decreasing over time.
II	Curvature	`4Λ′` — the curvature of the surface at the operating point; maximal at the inflection point.
Λ	Logistic function	The saturating function used in many channels.
H⋆	Threshold	The minimum level for the normal component to begin producing displacement (baseline 0.45).
‖[L_a,L_b]‖	Commutator	The degree of order-dependence of two influences.
H(γ)	Holonomy	The displacement remaining after a closed loop in parameter space.

7. Troubleshooting

Symptom	Resolution
The 3-D viewport is blank or black	The browser does not have WebGL or hardware acceleration enabled. Enable hardware acceleration in the browser settings and reload the page.
An *“out of range”* warning appears	The `τ` value (or a variable) has exceeded the usable region. This is an intentional notice, not an error; lower the value back into the region for the result to be valid.
Fonts look unusual	The network could not load fonts from the font server; the page falls back to system fonts, and the content is unchanged.
Interaction is slow or stutters	Shrink the browser window or close other tabs; 3-D rendering depends on the machine's graphics capability.